

Part III Seminar Abstracts - Michaelmas 2025

Algebra

Amy Needham

Thursday, 9:00am, MR12

The many products of groups

Given two groups (and possibly other data) there are many ways to combine them to form another group. A question arises of how much of the structure of the original groups are preserved by various kinds of product. In this talk we shall discuss what (if anything) is known about the residual finiteness and solubility of the word problem for various types of group products

David He

Thursday, 9:40am, MR12

Representation theory of finite monoids

Finite monoids have a rich representation theory that vastly generalises the situation of finite groups. I will present the fundamental result of Clifford–Munn–Ponizovskii which describes all simple representations of a finite monoid in terms of the simple representations of the monoid's maximal subgroups, and do many examples.

Algebraic Geometry

Avery Bainbridge

Thursday, 10:20am, MR12

Algebraic functions and differential equations

When do differential equations have algebraic solutions? This question has a rich history, with names like Gauss, Abel, Schwartz, and Klein attached, and some beautiful partial answers have resulted from their work. More recently, an arithmetic approach---the famous Grothendieck--Katz p-curvature conjecture---to this question has surfaced that boldly predicts that the only obstruction of the existence of algebraic solutions is a local one. In the first half of the talk I will motivate the original question and shed light on its algebraic-geometric interpretation. In the remaining time, I will outline the modern arithmetic approach to this question and discuss known cases where it has been successful.

Alex Karbowski

Thursday, 12:00pm, MR12

Poncelet's Theorem

Let C and D be two nested ellipses in the plane. When does there exist a polygon of n -sides inscribed in C and circumscribed about D ? Poncelet's Theorem is a result from classical projective geometry that provides an interesting answer to this question: there either exists none, or there exists an infinite family of such n -gons. We follow a 1977 proof of Poncelet's theorem by Griffiths and Harris, which reveals that the configuration of conics C and D can be associated with an elliptic curve. We finish by discussing how to detect exactly when such n -gon exists, a result first attributed to Cayley in 1853.

Yoyo Jiang

Thursday, 12:40pm, MR12

Espace étalé is a Kan extension

In this talk, we aim to give a conceptual presentation of the espace étalé (or étale space) construction in sheaf theory. We will begin with a quick reminder of what (and why) sheaves are, and in particular see a few examples of sheaves that come from taking sections of our favourite bundles. The rest of the talk is dedicated to convincing the audience that all sheaves are sections to

some bundle by constructing the espace étalé associated to a presheaf, which we will do by introducing Kan extensions and unpacking the formula for pointwise Kan extensions in this context. In particular, this will give us another way to conceptualize the sheafification functor. If time permits, we will point out a few other examples of Kan extensions that show up in basic sheaf theory in an attempt to convince the audience of their ubiquity.

Pawel Pielasa

Thursday, 1:20pm, MR12

Decompositions of Chow rings of direct sums of matroids

We prove two dual recursive decompositions as a graded $CH(M) \otimes CH(N)$ -module of the Chow ring $CH(M \oplus N)$ of the direct sum of matroids. We use this to obtain a decomposition of $CH(M \oplus N)$ into irreducible $CH(M) \otimes CH(N)$ -modules. The result implies a new recursive formula for the Eulerian numbers. Similarly, we find a recursive decomposition of the augmented Chow ring $CH(M \oplus N)$ into $CH(M) \otimes CH(N)$ -modules, generalizing some of the results of [BHMW22]. We prove analogous decompositions of (augmented) Chow polynomials of weakly ranked posets in the sense of [CF25].

Differential Geometry and Topology

Anthony Dender

Thursday, 3:40pm, MR13

Undergeodesics

Depending on one's context, there are various ways in which one could say that a space, or a curve within a space, is "close to" being geodesic: length spaces, quasi-geodesic curves, etc. However, none of these concepts have any meaningful application in a metric space that is highly non-path-connected (e.g. the rational numbers, viewed as a subspace of the reals), and in such cases the majority of techniques of metric geometry fail to give useful information. This talk will attempt to address this by introducing "undergeodesics", exploring their structure, and in particular showing a canonical way in which a space that is close to being geodesic in an "undergeodesic" sense can be embedded into an associated geodesic space.

Ariff Jazlan Johan

Thursday, 4:20am, MR13

Hochschild homology and string topology

String topology is the recently fashionable preoccupation with the homology of free loop space. It has deep geometric roots in the search for closed geodesics (via Morse theory on free loop space) and in Sullivan's program to understand the "algebraic... meaning of a... manifold". Considerable effort has been made since its inception in 1999 to uncover the vast zoo of (co)operations that arise in this context, and much of this is owed to the interpretation of string homology in terms of the Hochschild homology: the 'canonical' homology theory one attaches to the category of associative algebras. I will talk about the many beautiful ways to intuit this interpretation and relate it to recent advances in string topology.

Brian O'Callaghan

Thursday, 5:00pm, MR13

Formality (and Spheres)

Spheres are one of the simplest objects in topology. However, in string topology it turns out looking at even S^2 that there are two reasonable algebraic structures we can impose. This is due to issues arising with formality, a notion we will explain. For this we conduct a whistlestop tour of some important operads, explain what formality with respect to certain operads mean, and why exactly this fails in our case. This means we can at least give a choice of which of these is morally more correct

Max Silin

Thursday, 10:20am, MR14

Elliptic Curves over the Reals - Groups, Lattice Symmetries, and Riemann Surfaces
Playing with Elliptic Curves, Doubly Periodic Functions, and Loops.

Number Theory

Andrew Liu

Thursday, 9:00am, MR14

Berkovich Spectrum and Variation of Ostrowski's theorem

We will start by discussing the (new) construction of Berkovich Spectrum by Scholze with a short application for motivation. Then we continue with the key observation on the correspondence between seminorm on commutative ring and absolute value on Banach fields, and discuss variations of Ostrowski's theorem for further classification of absolute values.

Donald Li

Thursday, 9:40am, MR14

A Brief overview of Local Langlands

The Langlands Program is central to modern number theory, where the set of conjectures as a bridge between the representation theory of algebraic groups and Galois groups of field extensions. In this talk we start with an overview of Artin reciprocity, and explore some major results in Local Langlands Conjectures.

Particle Physics, Quantum Fields and Strings

Vatsal

Thursday, 9:00am, MR20

Fermionic Chern–Simons theory on $S^2 \times S^1$ in the ‘temporal’ gauge

Most of the computational evidence for the Bose–Fermi duality of fundamental fields coupled to $U(N)$ Chern–Simons theories originates in the large- N calculations performed in the light-cone gauge. The evaluation of the thermal free energy on a finite-sized sphere is elusive in the light-cone gauge but more natural in another gauge, the ‘temporal’ gauge. We use it to evaluate the finite-temperature partition function of $U(N)$ and $SU(N)$ coupled fermions at large N and finite chemical potential on S^2 , a novel physical setting for the problem. We set up the finite-temperature gap equations, solve them, and evaluate the partition function, closely following the tricks explored in [arXiv:2307.11020], where we work out the results on R^2 at large N , which demonstrate perfect agreement with the earlier lightcone gauge results. We find new formulae for the dynamically generated and flux-dependent masses (a completely novel feature with no flat-space counterpart). We look forward to announcing the second installment of our work soon.

Gary Louw

Thursday, 9:40am, MR20

Bose condensation in gluon plasma

We investigate gluon condensation in heavy-ion collisions by solving a relativistic Boltzmann equation for glasma-inspired initial conditions. We treat the standard case of a homogeneous and isotropic fluid and generalise an existing numerical approach for binary interactions in the scalar ϕ^4 -theory to QCD. Our scheme improves on existing test particle simulations and leading-log approximations of the collision term. Comparison of the latter to our model fixes an auxiliary UV cut-off which hitherto was known only parametrically. Uncertainties in the evolution are thereby reduced to details of QCD screening, which we parametrise with a screening mass arising from well-known HTL calculations. Preliminary results suggest that condensation times are underestimated by leading-log approaches.

Ansar Adibay

Thursday, 11:00am, MR20

Fundamentals of Supergeometry

Fundamental objects of supergeometry—such as superalgebras, superspaces, and supermanifolds—are introduced. Several important examples of supermanifolds are discussed, including the supersphere, super projective spaces, and super Grassmannians. The supergeometric language is then developed, and its applications to classical geometry are demonstrated; in particular, a slightly alternative proof of the Poincaré Lemma is presented.

Nathan Godechal

Thursday, 11:40am, MR20

Higher-form (or generalized) symmetries in QFT

Introducing the generalized definition of symmetries in QFT. Example in Maxwell U(1) or Chern-Simons. Discussion of phases and topological aspects of these symmetries.

Lorenz Hartmann

Thursday, 2:20pm, MR20

Why the universe would collapse without Projective Representations

Even though the rotation group $SO(3)$ only admits representations with integer spins $j=0,1,2,\dots$, we see half-integer spins in nature everywhere we look. This talk explains how projective representations come to the rescue and resolve this apparent contradiction.

Herbie Warner

Thursday, 3:00pm, MR20

Holographic Correlation Measures

Holography is the wild idea that spacetime geometry is a readable print out of quantum correlation: draw the right surface in the bulk, and you quickly learn something sharp about the entanglement structure of an otherwise intractable QFT state. This talk introduces that geometric dictionary starting from holographic entanglement entropy through the Ryu-Takayanagi formula, into the richer—and much harder—realm of mixed state correlations through entanglement wedge cross sections. We demonstrate how these “geometrised” correlation measures imply information theoretic inequalities on the underlying QFT and how one goes about proving these results in arbitrary dimensional AdS/CFT.

Applied and Computational analysis

Autumn Mapes

Thursday, 4:20pm, MR20

Designing video games that inspire mathematical reasoning

The medium of interactive video games are relatively unexplored as a means of math communication. As the designer of an upcoming math-inspired puzzle game, Threadbound, I'd like to share how the game connects to real world dynamical systems and how it reframes game engines as interactive mathematical simulations.

Combinatorics

Tomáš Flídr

Thursday, 3:30pm, MR2

Cops, robbers and ordinals

I will be talking about a simple version of the game of Cops and Robbers: two players move on the vertices of a graph one at a time, with the cop trying to catch the robber. After a brief discussion of finite graphs, we will move on to results on infinite graphs from my summer research project: we will show the existence of a graph where the capture time (length of the game) for any two initial

positions is finite, but their set is unbounded. We will also generalise the notion of a capture time to ordinals and show that for any ordinal α , there is a graph with capture time α .

Jude Horner

Thursday, 4:10pm, MR2

Nim on Towers of Hanoi

Tackling Project Euler Problem 806, an interesting problem based on the simple mathematical games of "Nim" and "Towers of Hanoi"

Probability

Domitille Saliou

Thursday, 4:50am, MR2

The characteristic polynomial of the Circular Unitary Ensemble

I will try to introduce some Random Matrix Theory, meaning defining the Circular Unitary Ensemble, and to draw parallels with the Riemann Zeta function.

Analysis and PDEs

Zach Halberstam

Friday, 9:00am, MR14

Geometry of PCF points in the Mandelbrot set

The Mandelbrot set is the set of points c such that the forward trajectory of 0 under iteration of the function $f_c(z) = z^2 + c$ remains bounded. The post-critically finite (PCF) parameters, those c such that this trajectory is finite, i.e., is eventually periodic, are of particular importance. In recent work DeMarco and Mavraki proved a number of sparsity results about varieties of degree d passing through many points in $\mathbb{C}^2$, both of whose coordinates are PCF parameters. In particular, they show there are finite many lines in $\mathbb{C}^2$ passing through three or more of these points, outside of trivial counterexamples. I discuss the importance of these results and recent efforts to understand the finite class of lines.

Phuoc Truong

Friday, 9:40am, MR14

A brief introduction to Harmonic Analysis

Maximal functions, Stein's maximal principle, Muckenhoupt maximal function (if time permits)

James Corcoran

Friday, 10:20am, MR14

(TBD) A Tourist's Guide to Gibbs Measures

Gibbs measures are the central object of equilibrium statistical mechanics and have also become a central interest to mathematicians working in probability and analysis. This talk aims to give a friendly non-technical overview on the origins of Gibbs distributions in mathematical physics, notable examples from daily life, and, if time permits, some more recent problems researchers have been interested in. The prerequisite knowledge is intended to be quite low to make the talk as accessible (and digestible) as possible.

Relativity and Cosmology

Kai Hall

Thursday, 1:40pm, MR20

Einstein-Cartan theory and Yang Mills geometry

Riemann-Cartan geometry can be viewed as a platform to reformulate GR as a gauge theory of local Lorentz symmetry. This makes the theory look much more like the Yang-Mills theories of the

standard model. There is a deep geometric structure underlying both curved spacetimes and gauge theories that will be explored in this talk. Einstein Cartan theory couples torsion to spinors and in principal differs from standard GR in its physical predictions, albeit for Planck scale physics.

Lingyun Fu

Friday, 9:00am, MR20

Quantum Treatment of Bound-state Black Hole Superradiance
TBA

Thomas Brosnan

Friday, 9:40am, MR20

Dynamical Dark Energy from Causal Set Theory

Dark energy is the name given to the unknown source driving the accelerated expansion of our universe. it constitutes 70% of the energy density of our universe, yet we have no idea what it is. Causal Set Theory is an approach to quantum gravity, that posits space and time to be fundamentally discrete. This talk will introduce how the fundamental principles of Causal Set Theory can be used to arrive at a natural explanation for dark energy that fluctuates over time and hence is dynamical. This has the potential to ease several of the tensions and problems with the cosmological constant. `

Elias Fink

Friday, 12:30pm, MR20

Cross-correlating long- and short-distance limits for a stochastic GW background

Gravitational Waves play an increasingly important role in astronomy - extending the capabilities of traditional light astronomy. The Stochastic Gravitational Waves Background is the superposition of gravitational waves from many uncorrelated cosmological and astrophysical sources. I will discuss experimental and theoretical approaches for isolating and measuring this as well as a novel approach developed over this summer. See more information at [arXiv:2510.25646](https://arxiv.org/abs/2510.25646).

Eric Lizalde

Friday, 13:10am, MR20

Searching for low-frequency GWs through the libration motion of the Moon and other satellites

Gravitational waves are a new window to explore the universe, and thus to new physics. However, current GWs detections such as the LIGO-VIRGO-KAGRA experiment only covers a small part of the spectra. Very recently, there has been a promising signal of GWs in the nHz band (from Pulsar Timing Arrays), of potentially a population of supermassive black hole binaries. In this work I explored the effect of gravitational waves in the muHz band in the rotational dynamics of the Moon, and other tidally locked satellites. This is a promising astrophysical system since we can measure very precisely the position of the Moon (up to cm!) with Lunar Laser Ranging experiments.

Quantum Computation, Information and Foundations

Ivan Shum

Thursday, 3:40pm, MR20

Quantum Gibbs Sampling of Stabilizer Hamiltonians

The notion of quantum Gibbs sampling will be introduced, explicit algorithms for stabilizer Hamiltonians will be derived, and implications for the usefulness of quantum computers will be discussed.

Continuum Mechanics

Pehan Amarathunge

Friday, 10:20am, MR20

Viscoelastic Instabilities: Instability without Inertia

Viscoelastic flows are prone to novel instabilities that arise due to elasticity alone, or due to a combination of elastic and inertial effects, and many such instabilities have no evident Newtonian counterparts. An understanding of purely elastic instabilities is of paramount importance to the plastics industry to prevent their occurrence during polymer processing operations, such as extrusion. In this talk, we will survey viscoelastic instabilities in various complex geometries and explain how the underlying microstructure can be used to explain the mechanism that drives these instabilities.